MCQS OF BT301(BIOTECHNOLOGY) LESSONS (96-209)

EFFORT BY

MARIA KANWAL

VU ZOOLOGISTS GROUP

MSc. ZOOLOGY 2ND SEMESTER.

Virtual University of Pakistan

BT301 Introduction to Biotechnology

2nd Semester MSc Zoology 2016-18

MCQs for Finals

(Topics 96-209)

VU ZOOLOGISTS

1- pBR322 is an example of in _____ constructed cloning vehicle.
- Intro
- Extro
- **Vitro**
- In-extro

2- It is difficult to clone DNA segments longer than about _____ when plasmids are used as the vector
- 10,000 bp
- **15,000 bp**
- 20,000 bp
- 25,000 bp

3- Plasmid _____ has a number of characteristics that make it suitable as cloning vehicle.
- **pBR322**
- pBS322
- pRR322
- pBR232

4- Over the years numerous different derivatives of _____ have been constructed.
- pBR232
- pRR322
- **pBR322**

- pBS322

5- pUC19 carrying sites for many different _____ endonucleases
- addition
- active
- moving
- **restriction**

6- Charon 4A and 16 Each has the lacZ gene, encoding enzyme βgalactosidase .

7- Connecting the two lambda fragments to foreign DNA by using _____ .
- **DNA ligase**
- Polymarise
- RNA
- Polymarise II

8- Packaging in vitro yields about _____ of vector DNA

- 104 plaques/ µg
- 105 plaques/ µg
- **106 plaques/ µg**
- 107 plaques/ µg

9- Plasmids have been constructed which contain a fragment of ʎ DNA including _____ .

- Sin site
- **cos site**
- tan site

10- Cosmids are particularly attractive vectors for constructing libraries of _____ fragments .

- **eukaryotic genomic**

VU ZOOLOGISTS

- prokaryotic genomic
- phynotype

11- Modified schemes for cloning in Cosmid vectors devised by Ish-Horowicz and Burke in _____

- 1980
- **1981**
- 1982
- 1983

12- Phasmid vectors are _____ of plasmid and ʎ phage sequences .

- Separation
- **Combination**
- Addition
- Davison

13- Bacterial artificial chromosomes are simply plasmids designed for the cloning of very long segments of DNA _____.

- 200,000 to 300,000 bp
- **100,000 to 300,000 bp**
- 150,000 to 400,000 bp
- 160,000 to 350,000 bp

14- Yeast artificial chromosomes have sites where very _____ fragments can be inserted

- Large RNA
- Messenger RNA
- **Large DNA**
- Small DNA

15- YACs with inserts more than _____ are nearly as stable as normal cellular chromosomes

- 140,000 bp
- 130,000 bp

VU ZOOLOGISTS

- 120,000 bp
- **150,000 bp**

16- Vectors that can replicate and are stably maintained in two or more unrelated host organisms are called shuttle vectors

- **shuttle vectors**
- clone vectors
- double vctors
- vaccines

17- Any cell based cloning procedure has _____ essential parts

- three
- **four**
- five
- six

18- Genome size (kb) of human is _____.

- 1.8 x 10^6
- **2.8 x 10^6**
- 3.8 x 10^6
- 2.8 x 10^6

19- PCR with specific primers could be used to _____.

- **isolate genes**
- break genes
- remove genes

20- High molecular weight genomic DNA is digested with _____ .

- pBR322
- **Sau3AI.**

21- cDNA is prepared by reverse-transcribing cellular _____ .

VU ZOOLOGISTS

- DNA
- **RNA**

22- _____ are rare in bacteria but occur in genes of higher eukaryotes

- **Introns**
- Exrons

23- _____ is a sensitive means for detecting, quantifying and cloning specific cDNA molecules

- ST-PCR
- **RT-PCR**

24- Plasmid and _____ vectors carry drug resistance or nutritional marker

- DNA
- **cosmid**

25- Grunstein and Hogness developed a screening procedure in _____.

- 1974
- **1975**
- 1976
- 1977

26- Screening by hybridization can be monitored by _____ .

- **Autoradiography**
- Radiology
- Hybridization

27- A _____ genomic fragment must be found as a starting point for the walk .

- Vector
- **Cloned**

VU ZOOLOGISTS

- Clear

28- Chromosome jumping involving a series of _____ manipulations.

- Chemical
- Biological
- **Biochemical**

29- Chromosome jumping is in the cloning of human _____ gene.

- **cystic fibrosis**

30- The PCR is widely used to isolate specific DNA sequences from _____ genomic DNA .

- **uncloned**

31- A _____ is a mixture of primers, all of similar sequences but with variations at one or more positions

- **degenerate primer**

32- Expression libraries are useful because they allow a range of _____ to be employed

- **alternative techniques**

33-Immunochemical screening involves the use of _____ .

- **antibodies**

34- In Screening expression libraries the molecular target for recognition is generally an _____ .

- **epitope**

VU ZOOLOGISTS

35- Immunochemical screening of ʎgt11 is much more convenient to use _____ insertion vectors.

- **bacteriophage-ʎ**

36- The original detection method using iodinated _____ .

- **antibodies**

37- Screening is carried out without using an antibody, by incubating the membranes with _____ double stranded DNA probe

- radiolabelled

38- Functional complementation is the process by which a particular DNA sequence compensates for a missing function in a _____.

- **mutant cell**

39- Functional complementation has been used for complementation in _____ mice to isolate the Shaker- 2 gene

- **transgenic**

40- _____ bacteria such as E. coli have a complex wall-membrane structure

- **Gram-negative**

41- All of the proteins are first synthesized in the _____ but somehow find their way out .

- **cytoplasm**

42- Proteins trafficking are actually several general mechanisms of protein export in _____ .

- **bacteria**

VU ZOOLOGISTS

43- Expression from a strong promoter can represent _____ of cloned gene product of toral cell protein

- **2040%**

44- Complementarity of Shine-Dalgarno (S-D) sequence with _____ can affect the rate of translation .

- **16s rRNA**

45- Composition of triplet immediately preceding the AUG start codon also affects the efficiency of _____ .

- **translation**

46- _____ of mRNA usually proceeds by a combination of endonuclease and 3'exonuclease attack.

- **Degradation**

47- One way of increasing the expression of a cloned gene is to _____ the number of the corresponding transcript .

- **Increase**

48- The loss of plasmids due to defective partitioning is called _____ .

- **segregative instability**

49- Structural instability of plasmids may arise by _____ or rearrangements of DNA .

- **deletions**

50- Factors which will be important include the choice of nutrients and _____ parameters .

- **environmental**

VU ZOOLOGISTS

51- Host cell _____ can affect the level of expression .

- **physiology**

52- The number of _____ in a cell far exceeds any one class of mRNA.

- **ribosomes**

53- The loss of plasmids due to defective partitioning is called _____ .

- **segregative instability**

54- DNA sequence information is a prerequisite for planning any _____ manipulation of the DNA .

- **substantial**

55- Techniques for DNA sequencing became available in the late _____.

- **1970s**

56- _____ method method for DNA sequencing makes use of chemical reagents to bring about base-specific cleavage of the DNA .

- **Maxam-Gilbert**

57- Chain termination or dideoxy procedure method was developed by Sanger et al. _____ .

- **1977**

58- Improvements to original Sanger's method have been made by replacing the _____ of E. coli DNA polymerase I .

- **Klenow fragment**

VU ZOOLOGISTS

59- Sequences that were read beyond 4oo bp contained an average of _____

- **3.2% error**

60- Sequences that were read less than 4oo bp had _____ .

- **2.8% error**

61- _____ is a process to change the genetic information of an organism .

- **Mutagenesis**

62- There are _____ different methods of site-directed mutagenesis have been devised .

- **Three**

63- A synthetic DNA fragment containing the desired mutant sequence is used to replace the corresponding sequence in the _____ .

- **wild-type gene**

64- After E. coli transformation, _____ that are formed are either that of original wild type DNA or that containing the mutated base .

- **heteroduplex**

65- _____ for site directed mutagenesis single bases mismatched between the amplification primer .

- **PCR methods**

66- The advantage of a PCR based mutagenic protocol is that the desired mutation is obtained with _____ efficiency

VU ZOOLOGISTS

- **100%**

67- PCR is a technology in molecular biology used to _____ a single or a few copies of a target DNA.

- **amplify**

68- PCR can also be used to generate DNA fragments for_____.

- **gene cloning**

69- _____ are enzymes that synthesize DNA molecules by adding nucleotide, the building blocks of DNA

- **DNA polymerases**

70- thermostable DNA polymerase isolated from thermophilic bacterium is _____ .

- **taq polymerase.**

71- Primer is a _____ of oligonucleotide that serves as a starting point for DNA synthesis .

- **short strand**

72- a mixture of primers, all of similar sequence but with variations at one or more positions is called as _____.

- **Degenerate primer**

73- Various strategies can be adopted for first-strand_____.

- **cDNA synthesis**

74- One use of RT-PCR is in determining the _____ in a sample .

VU ZOOLOGISTS

- **amount of mRNA**

75- qPCR is used to measure the quantity of a_____.

- **target sequence**

76- _____ is used to measure starting amounts of DNA, cDNA, or RNA .

- **Real-time quantitative PCR**

77- In _____ , two sets of primers are used in two successive PCRs .

- **Nested PCR**

78- _____ is used to identify the flanking sequences around genomic inserts .

- **Inverse PCR**

79- _____ is used to amplify several different DNA sequences simultaneously.

- **Multiplex PCR**

80- RAPD stands for _____ .

- **Random amplification of polymorphic DNA**

81- In RAPD the segments of DNA that are _____ are random .

- **amplified**

82- RFLP stands for _____ .

- **Restriction fragment length polymorphism**

VU ZOOLOGISTS

83- RFLP exploits variations in _____ sequences .

- **Homologous DNA**

84- In RFLP, DNA sample is digested by _____ .

- **restriction enzymes**

85- In RFLP, DNA sample is separated by _____ .

- **gel electrophoresis**

86- AFLPs stands for _____ '

- **Amplified fragment length polymorphisms**

87- AFLPs are differences in restriction fragments length caused by _____ .

- **SNPs or INDELS**

88- AFLP is a PCR based tool used in _____.

- **genetic engineering**

89- PCR has been used in gene cloning and screening of _____.

- **genomic libraries**

90- PCR has major impact on medicine especially in the field of _____ or diagnosis .

- **clinical microbiology**

91- The application of scientific procedures to solve criminal and legal matters is called _____ .

VU ZOOLOGISTS

- **Forensic science**

92- To obtain evidence from minimal samples of saliva, semen or other tissues include in _____ .

- **Molecular methods**

93- Fingerprinting is a forensic technique used to identify individuals by characteristics of their _____ .

- **DNA**

94- PCR has facilitated research in detection of _____ in plants, animals and environment

- **Pathogens**

95- The recovery and analysis of DNA and protein from ancient human, animal and plant remains is called _____.

- **Molecular paleontology**

96- Assigning of a specific gene to particular region of a chromosome and determining the location of and relative distances between genes on the chromosome called as _____ .

- **Genome mapping**

97- The landmarks on genome map might include_____ , regulatory sites and genes themselves .

- **short DNA sequences**

98- Genetic linkage maps illustrate the order of _____ .

- **Genes**

VU ZOOLOGISTS

99- _____ give the DNA base pair distances from one landmark to another

- **Physical maps**.

100- _____ of bacteria has been obtained by combining the mapping techniques .

- **Chromosomal maps**

101- Botstein et al _____ were the first to recognize that DNA probes that target RFLPs can be used for mapping

- **1980**

102- RFLP-Restriction fragment length polymorphism exploits variations in _____ sequences

- **homologous DNA**

103- STS stands for _____

- **Sequence-tagged sites**

104- (STS) are more convenient markers than _____ because they do not use southern blotting

- **RFLPs**

105- Single nucleotide polymorphisms (SNPs) are different sequence alternatives (alleles) exist in a _____.

- **population**

106- SNPs probably are the most important sequence markers for _____ of genomes

VU ZOOLOGISTS

- **physical mapping**

107- Polymorphic DNA can be detected by _____.

- **amplification**

108- Amplified fragment length polymorphisms (AFLP) is a diagnostic _____.

- **fingerprinting technique**

109- FISH stands for _____ .

- **Fluorescence in situ hybridization**

110- Fluorescent probes that bind to only those parts of the chromosomes with a _____ of sequence complementarity .

- **high degree**

111- RH mapping used _____ breakage of chromosomes

- **X-ray**

112- RH mapping used X-ray to determine the_____.

- **order on the chromosome**

113- _____ overcomes these limitations, being an entirely in vitro technique.

- **Happy mapping**

114- The different mapping methods are integrated as each has its own strengths and weaknesses .

- **Correct statement**

VU ZOOLOGISTS

115- The process of determining the exact order of nucleotides within a DNA or RNA molecule is called _____ .

- **sequencing**

116- In_____ , DNA sequencing was a very tedious process which involved determining only a few hundred nucleotides at a time .

- **1970s**

117- In _____ strategy, DNA shredded into smaller fragments

- **Shotgun sequencing**

118- Shotgun sequencing is used for sequencing _____ .

- **long DNA strands**

119- a map of each chromosome of the genome is made before the DNA is split up into fragments for sequencing is called _____ .

- **Clone-by-clone sequencing**

120- _____ are homologous genes in different organisms

- **Orthologs**

121- Genome of bacteria may be of variable in size from _____ .

- **0.49 Mb to 9.1 Mb**

122- _____ genomes exhibit an amazing structural diversity .

- **Mitochondrial**

VU ZOOLOGISTS

123- Gene transfer has occurred between _____ and nuclear DNA .

- **mtDNA**

124- The minimal _____ is smaller than many bacterial genomes .

- **eukaryotic genome**

125- _____ genomics of eukaryotes gives insight into the evolution of key proteins .

- **Comparative**

126- _____ are used to study the expression of many genes at once.

- **Microarrays**

127- Spotted DNA arrays are produced by printing DNA samples on treated microscope slides .

- **Correct statement**

128- The predominate application of DNA microarray has been to measure _____ .

- **gene expression levels**

129- Microarrays have been widely used as _____ genotyping platforms .

- **single nucleotide polymorphism (SNP)**

130 - Phage display is the technology that allows expression of _____ on the surface of phage particles .

- **exogenous polypeptides**

VU ZOOLOGISTS

131- the phage clones with binding affinity for the target by a process called _____ .

- **biopanning**

132- A genetic technique in which one of an organism's gene is made inoperative that is simply called as_____.

- **knocked out**

133- the insertion of a protein coding cDNA sequence at a particular locus in an organism's chromosome is called _____ .

- **knock-in**

134- siRNA is the method of choice to target specific genes for _____ .

- **silencing**

135- Oligonucleotide chips are manufactured by in _____ synthesis.

- **situ oligonucleotide**

VU ZOOLOGISTS

Good luck ☺